Loan Approval Dataset
Exploratory Data Analysis

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DATA OVERVIEW

The dataset offers a detailed view of loan applications with key financial and demographic features that contribute to assessing loan approval or rejection:

📝 Feature Descriptions:

Column	Description	Data Type
person_age	Applicant's age	Float
person_gender	Applicant's gender	Categorical
person_education	Applicant's highest level of education	Categorical
person_income	Applicant's annual income in currency	Float
person_emp_exp	Years of employment experience	Integer
person_home_ownership	Home ownership status (e.g., rent, own, mortgage)	Categorical
loan_amnt	Amount of loan requested	Float
loan_intent	Intended purpose of the loan (e.g., personal, education)	Categorical
loan_int_rate	Interest rate applicable to the loan	Float
loan_percent_income	Loan amount as a percentage of annual income	Float
cb_person_cred_hist_length	Number of years of credit history	Float
credit_score	Applicant's credit score	Integer
previous_loan_defaults_on_file	Indicator of previous loan defaults (Yes/No)	Categorical
loan_status	Loan status outcome (1 = approved, 0 = rejected)	Integer

In the real world credit-scoring:

- ✓ Credit history, income stability and loan-to-income ratios are major approval fact ors.
- ✓ Having a high credit score and longer cre dit history increase the loan approval cha nces
- ✓ Previous loan defaults and high-risk loan purposes (e.g., venture loans) often lead t o rejection.
- ✓ Loan percent income is crucial in determ ining if a loan is affordable.
- ✓ Interest rates may reflect risk-based prici ng — higher rates might actually increas e approval if the lender can price in the ri sk.

DATA OVERVIEW (DOMAIN REFERENCES)

	SOURCE	SUMMARY	REFERENCES
Credit History, Defaults & Credit Scoring	FICO Credit Score System	Payment history, credit length, and default records are the most influential factors in credit scoring.	Fair Isaac Corporation. (2023). What's in my FICO® Score?
Income Stability & Loan-to-Income Ratios	OECD & World Bank Credit Risk Guidelines	A borrower's debt-to-income (DTI) or loan-to-income ratio is commonly used to assess loan affordability.	OECD (2020). Responsible Lending in the Digital Age. World Bank. (2021). Credit Reporting Knowledge Guide.
Interest Rates & Risk-Based Pricing	Federal Reserve Bank of St. Louis	Lenders often approve higher-risk borrowers by assigning higher interest rates to compensate for credit risk.	Board of Governors of the Federal Reserve System. (2022). Risk-Based Pricing
Loan Purpose and Approval Likelihood	Consumer Financial Protection Bureau (CFPB)	Certain loan purposes (e.g., debt consolidation, home improvement) are considered safer than others like startup or venture loans.	CFPB. (2022). Consumer Credit Trends Report cfpb.gov/data-research
Fair Lending & Underwriting Automation	European Banking Authority (EBA) & IMF Reports	Automating approvals improves consistency, but domain-informed rules (like default history or affordability) are essential for fairness and regulatory compliance.	EBA (2022). *Guidelines on Loan Origination and Monitoring* IMF (2020). *The Role of Big Data and Machine Learning in Credit Scoring*

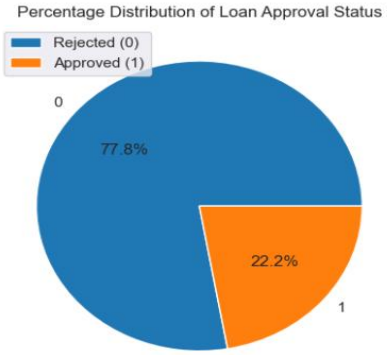
STATISTICAL SUMMARY

8]:

	count	mean	std	min	25%	50%	75%	max
person_age	45000.0	27.764178	6.045108	20.00	24.00	26.00	30.00	144.00
person_income	45000.0	80319.053222	80422.498632	8000.00	47204.00	67048.00	95789.25	7200766.00
person_emp_exp	45000.0	5.410333	6.063532	0.00	1.00	4.00	8.00	125.00
loan_amnt	45000.0	9583.157556	6314.886691	500.00	5000.00	8000.00	12237.25	35000.00
loan_int_rate	45000.0	11.006606	2.978808	5.42	8.59	11.01	12.99	20.00
loan_percent_income	45000.0	0.139725	0.087212	0.00	0.07	0.12	0.19	0.66
cb_person_cred_hist_length	45000.0	5.867489	3.879702	2.00	3.00	4.00	8.00	30.00
credit_score	45000.0	632.608756	50.435865	390.00	601.00	640.00	670.00	850.00
loan_status	45000.0	0.222222	0.415744	0.00	0.00	0.00	0.00	1.00

Number of Records : 45,000
Total features : 14 (5 categorical, 9 numerical)

- ✓ Age ranges 20 to 144 (potential outliers)
- ✓ The mean income is ~80k but the maximum is ~720k indicating possible outliers
- ✓ Loan amount has median of ~8k with a max value of ~35k
- ✓ The credit score has a range of 390-850

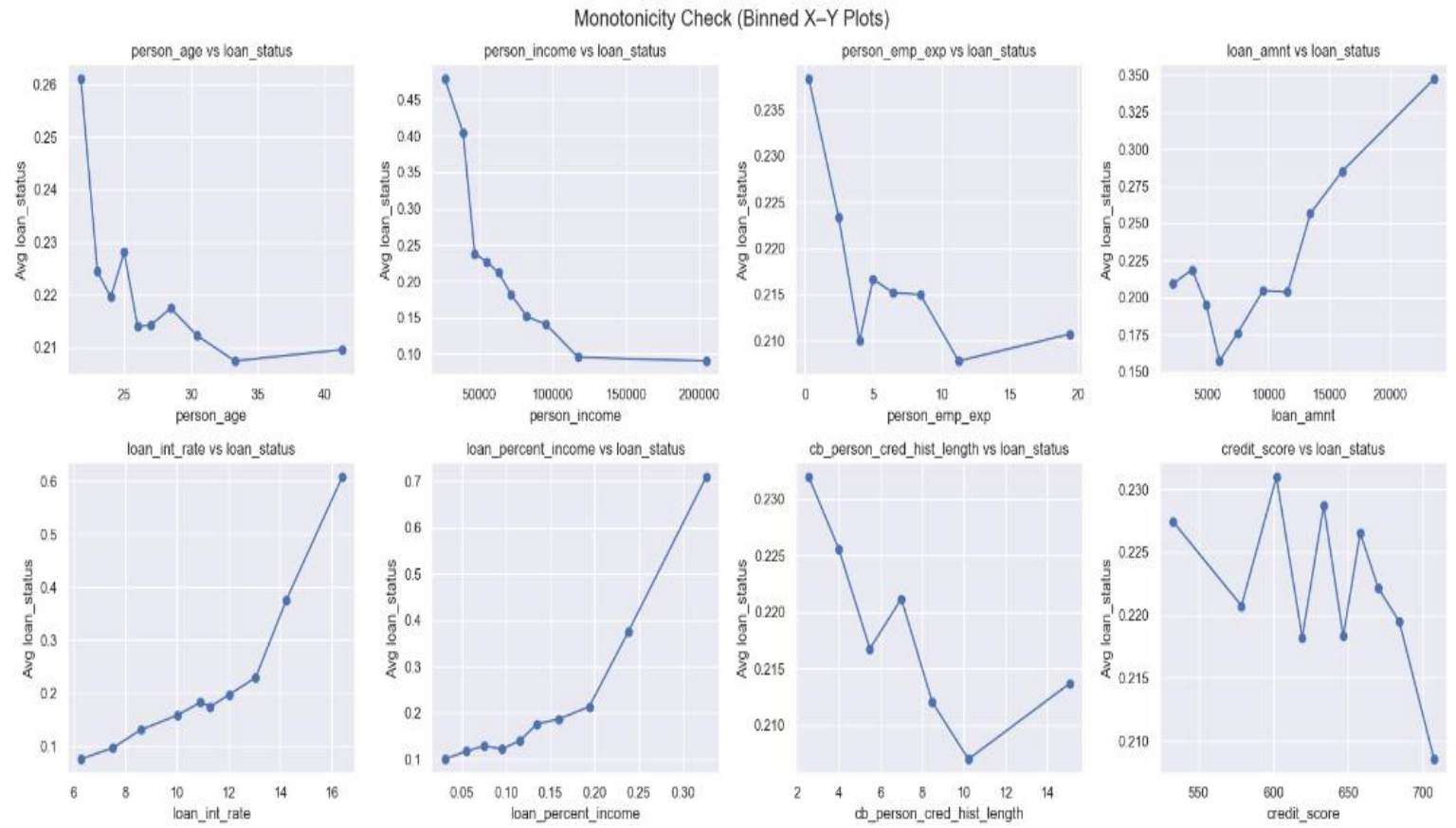


- ✓ The loan status indicates that approximately ~22% of the loans were approved

KEY INSIGHTS – SPEARMAN CORRELATION

Feature	Correlation	Interpretation
loan_percent_income	+0.324	• Unexpected positive monotonic – higher loan burden is linked to higher approval. • **Applicants with larger loan ratios may be pre-screened or higher quality in other ways.**
loan_int_rate	+0.311	• Might reflect **lender profitability incentives** or risk-based pricing.
loan_amnt	+0.087	• Very weak positive relationship — loan size doesn't strongly influence approval likelihood.
person_income	−0.275	• Higher income is linked to **slightly lower app roval**, possibly due to requesting larger loans or stricter underwriting. • May point to **non-linear effects** or interactio ns with other variables.
Others (person_age, person_emp_exp, cb_person_c red_hist_length, credit_score)	~0	• No clear monotonicity observed

KEY INSIGHTS – X-Y PLOTS



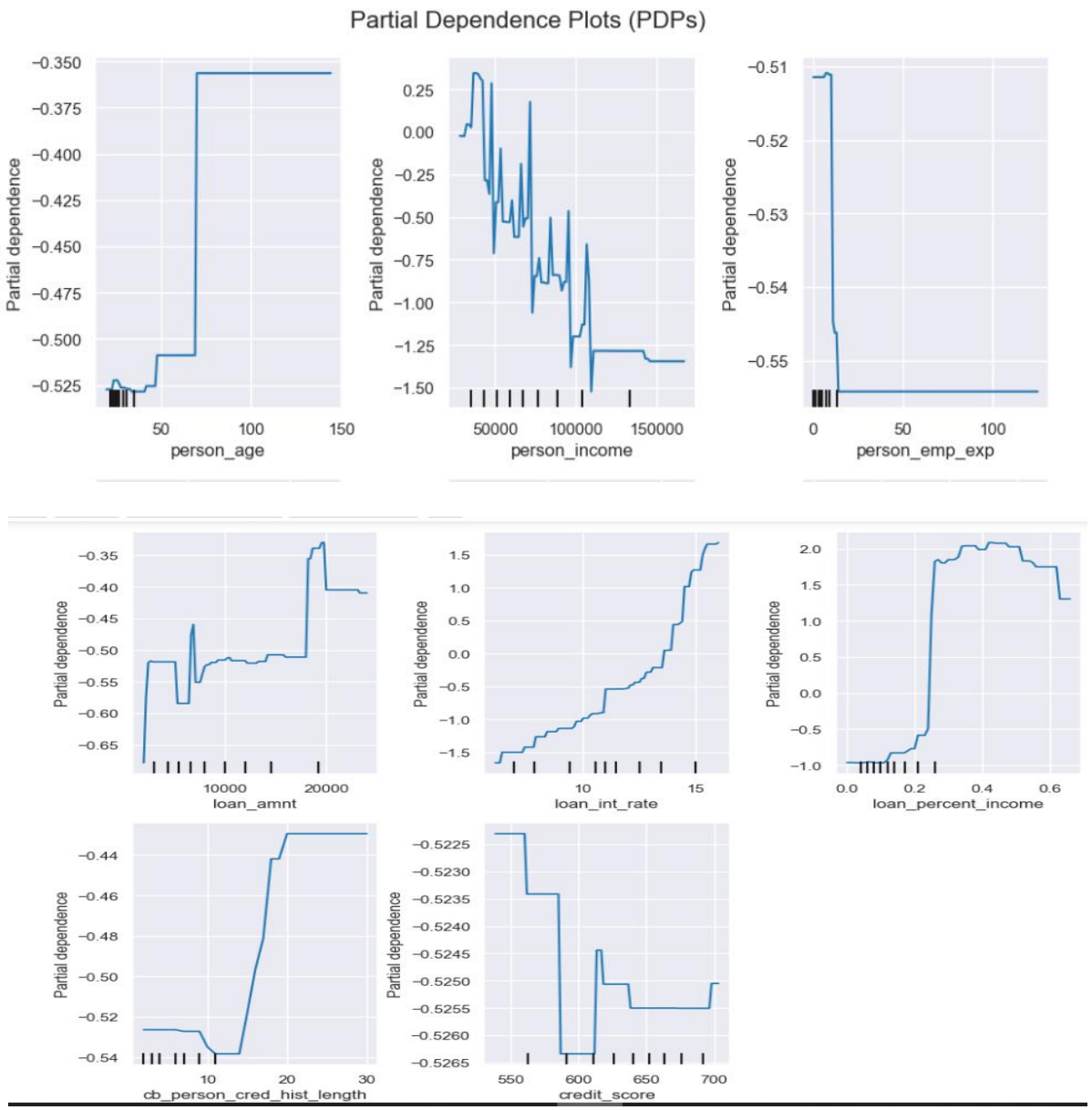
Negative Monotonic Trends:

- person_income, person_age, person_emp_exp
  - Younger/less experienced applicants might be receiving more approvals — perhaps due to loan types (e.g., student or startup).

Positive Monotonic Trends:

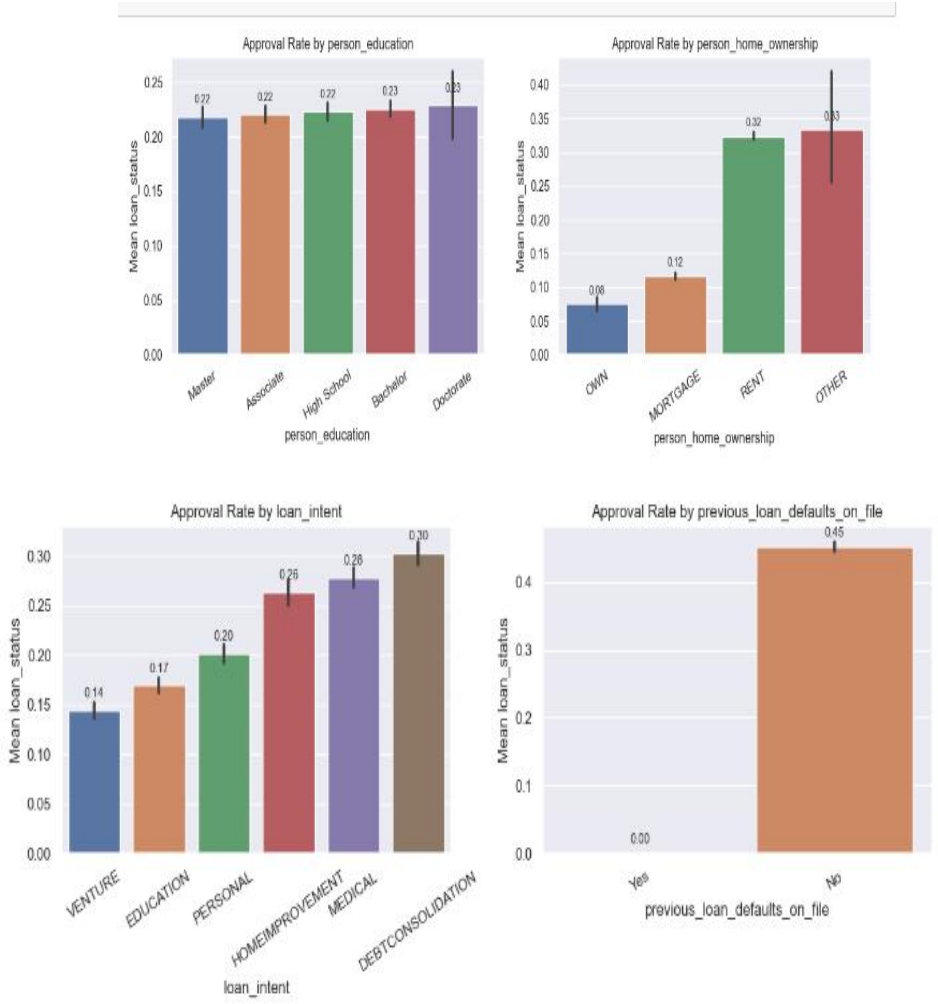
- loan_amnt
  - Larger loan amounts have **higher approval** – could indicate **confident or qualified borrowers** requesting more.
- loan_int_rate
  - Higher interest rates correlate with **higher approval** – may reflect **lender incentive** to approve high-profit loans.
- loan_percent_income
  - Surprisingly, higher loan burden (as % of income) leads to **higher approval** – may correlate with risk-based pricing or repayment potential.

KEY INSIGHTS – PDP PLOTS



- Higher income or older/more experienced applicants are not favored — suggests the model may be more favorable toward **younger or mid-income profiles**.
- Higher interest rates → more approvals (may reflect **lender profit** motive).
- Higher loan burden (loan/income ratio) → surprisingly **more approvals**, which may reflect a **targeted high-risk/high-reward segment**.
- Larger loans are more likely approved, possibly due to association with **more credit worthy or prioritized borrowers**

KEY INSIGHTS – CATEGORICAL FEATURES



- **Home Ownership**: **Renters** have **much higher approval rates** (~32%) than mortgage holders (~12%) — possibly due to lower debt burden
- **Loan Intent**:
  - **Debt consolidation / medical** → highest approval
  - **Venture loans** → lowest approval (~14%) ~ Riskier
- **Defaults on File**: **Zero approvals** for applicants with previous defaults → indicates **strict binary rule**

SUMMARY

- **Goal**: Explore loan approval patterns using applicant and loan data
- **Loan-related features** (interest rate, loan burden) show strong patterns with approval
- **Demographic features** (income, age, credit score) had weak or unexpected influence
- Applicants with **previous defaults** were **never approved** → strict historical rule
- **Non-linear relationships** observed → validates use of PDPs & Spearman

EDA shows that loan-related factors drive approval much more than demographic variables, and some patterns—like zero approval for defaulters—suggest rigid rules. We also observed non-linear trends that we'll account for when we move into modeling.

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